Geometry: Proving Triangle Similarity

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Name: _____ Date: _____ Score: / 10

DIRECTIONS *Use similarity postulates and theorems to solve each problem. Show all work.*

1 Identify the postulate shown:

$\triangle ABC: \angle A = 50°, \angle B = 60°$ $\triangle DEF: \angle D = 50°, \angle E = 60°$

Answer: _____

2 Find the missing angle in ΔPQR:

$\angle P = 45°, \angle Q = 75°, \angle R = ?$

Answer: _____

3 Are the triangles similar? State why:

$\triangle JKL: \angle J = 30°, \angle L = 70°$ $\triangle PQR: \angle Q = 30°, \angle R = 70°$

Answer: _____

4 Find ∠K, then check similarity:

$\triangle JKL: \angle J = 85°, \angle L = 70°$ $\triangle PQR: \angle P = 25°, \angle R = 70°$

Answer: _____

5 Explain why the triangles are similar:

$\angle 1$ and $\angle 2$ are vertical angles; $\angle R = \angle V = 45°$

Answer: _____

6 Find ∠P to confirm similarity:

$\triangle JKL: \angle J = 30°, \angle L = 70°$ $\triangle PQR: \angle Q = 85°, \angle R = 70°$

Answer: _____

7 Solve for x using AA similarity:

$\triangle ABC \sim \triangle DEF, \quad \angle A = (3x + 10)°, \angle D = 55°$

Answer: _____

8 Find x and y given similarity:

$\triangle RST \sim \triangle XYZ, \quad \angle R = (4x)°, \angle X = 60°, \angle S = (3y + 6)°, \angle Y = 48°$

Answer: _____

9 Prove similarity using all steps:

$\triangle ABC: \angle A = (2x + 5)°, \angle B = 80°$ $\triangle DEF: \angle D = 45°, \angle E = 80°$

Answer: _____

10 Find x, y, and write the similarity statement:

$\triangle PQR: \angle P = (5x - 10)°, \angle Q = (2y + 4)°, \angle R = 66°$ $\triangle STU: \angle S = 40°, \angle T = 74°, \angle U = 66°$

Answer: _____

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Answer Key & Solutions

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TEACHER NOTES *This worksheet reinforces the Angle-Angle (AA) Similarity Postulate, which states that two triangles are similar if two pairs of corresponding angles are congruent. Remind students that the sum of interior angles of a triangle equals 180°, which can be used to find missing angles and confirm similarity. Encourage students to write*

1 Identify the postulate shown:

$\triangle ABC: \angle A = 50°,\ \angle B = 60°\quad \triangle DEF: \angle D = 50°,\ \angle E = 60°$

= $\triangle ABC \sim \triangle DEF$

Two pairs of corresponding angles are congruent (∠A ≅ ∠D and ∠B ≅ ∠E), so the triangles are similar by the AA Similarity Postulate.

2 Find the missing angle in ΔPQR:

$\angle P = 45°,\ \angle Q = 75°,\ \angle R = ?$

= $\angle R = 180° - 45° - 75° = 60°$

The sum of interior angles of a triangle is 180°, so ∠R = 180° – 45° – 75° = 60°.

3 Are the triangles similar? State why:

$\triangle JKL: \angle J = 30°,\ \angle L = 70°\quad \triangle PQR: \angle Q = 30°,\ \angle R = 70°$

= $\triangle JKL \sim \triangle QRP$

∠J ≅ ∠Q and ∠L ≅ ∠R, giving two pairs of congruent corresponding angles, so the triangles are similar by AA.

4 Find ∠K, then check similarity:

$\triangle JKL: \angle J = 85°,\ \angle L = 70°\quad \triangle PQR: \angle P = 25°,\ \angle R = 70°$

= $\angle K = 180° - 85° - 70° = 25°$

∠K = 180° – 85° – 70° = 25°. Since ∠K ≅ ∠P = 25° and ∠L ≅ ∠R = 70°, the triangles are similar by AA.

5 Explain why the triangles are similar:

$\angle 1$ and $\angle 2$ are vertical angles; $\angle R = \angle V = 45°$

= $\triangle RSW \sim \triangle VSB$

Vertical angles are congruent, providing one pair; the given 45° angles provide a second pair, satisfying the AA Similarity Postulate.

6 Find ∠P to confirm similarity:

$\triangle JKL: \angle J = 30°,\ \angle L = 70°\quad \triangle PQR: \angle Q = 85°,\ \angle R = 70°$

= $\angle P = 180° - 85° - 70° = 25°$

∠K = 180° – 30° – 70° = 80° and ∠P = 180° – 85° – 70° = 25°. Since no two pairs of angles match, the triangles are not similar.

7 Solve for x using AA similarity:

$\triangle ABC \sim \triangle DEF,\quad \angle A = (3x + 10)°,\ \angle D = 55°$

= $3x + 10 = 55 \Rightarrow x = 15$

Since ∠A corresponds to ∠D and the triangles are similar, set 3x + 10 = 55, giving 3x = 45, so x = 15.

8 Find x and y given similarity:

$\triangle RST \sim \triangle XYZ,\quad \angle R = (4x)°,\ \angle X = 60°,\ \angle S = (3y + 6)°,\ \angle Y = 48°$

= $4x = 60 \Rightarrow x = 15;\quad 3y + 6 = 48 \Rightarrow y = 14$

Set 4x = 60 to get x = 15, and set 3y + 6 = 48 to get 3y = 42, so y = 14.

9 Prove similarity using all steps:

$\triangle ABC: \angle A = (2x + 5)°,\ \angle B = 80°\quad \triangle DEF: \angle D = 45°,\ \angle E = 80°$

= $2x + 5 = 45 \Rightarrow x = 20$

Set ∠A = ∠D: 2x + 5 = 45, so x = 20 and ∠A = 45°. Since ∠A ≅ ∠D and ∠B ≅ ∠E = 80°, the triangles are similar by AA.

10 Find x, y, and write the similarity statement:

$\triangle PQR: \angle P = (5x - 10)°,\ \angle Q = (2y + 4)°,\ \angle R = 66°\quad \triangle STU: \angle S = 40°,\ \angle T = 74°,\ \angle U = 66°$

= $5x - 10 = 40 \Rightarrow x = 10;\quad 2y + 4 = 74 \Rightarrow y = 35$

Set ∠P = ∠S: 5x – 10 = 40, so x = 10. Set ∠Q = ∠T: 2y + 4 = 74, so y = 35. With ∠R ≅ ∠U = 66° and ∠P ≅ ∠S = 40°, the triangles are similar by AA: ΔPQR ~ ΔSTU.

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